Client Class Diagram — The client-side MVC and network layer: ClientController (TUI and GUI variants), the LobbyModel/GameModel observed by ClientView, the ClientNetworkDispatcher, the Socket and RMI ServerProxy implementations, and the Event hierarchy with Inversion-of-Control dispatch. It shows how user actions are sent to the server and how incoming server events update the local models and views.

LEGEND

Stereotypes:

«interface»	interface
«abstract»	abstract class
«enumeration»	enum
«record»	record (immutable components)
«singleton»	single shared instance
«factory» / «utility»	role of the class
«ghost»	class defined in ANOTHER diagram or package; drawn empty (name only) as a target for arrows

Visibility: + public - private # protected ~ package-private

Modifiers: underlined = static italic = abstract {readOnly} = final

Relationships (line / head):

——▷ Generalization	SOLID line + hollow triangle	"extends"
- - -▷ Realization	DASHED line + hollow triangle	"implements"
◆—— Composition	filled diamond	whole owns the part (same lifetime)
◇—— Aggregation	hollow diamond	whole references the part (separate lifetimes)
——> Association	open arrow	stable reference (field)
- - -> Dependency	dashed open arrow	transient use (parameter / call)
⊕—— Nesting	inner (nested) class	

TuiView

- lobbyModel : LobbyModel
- gameModel : GameModel
- currentGameId : String
- notifications : LinkedList<String>
- eventQueue : BlockingQueue<Runnable>
- bannerPrinted : boolean
- pendingSummary : String
- MAX_NOTIFICATIONS : int

+ **TuiView**(lobbyModel : LobbyModel)
+ **setGameModel**(gameModel : GameModel) : void
- **startEventProcessor**() : void
- **addNotification**(message : String) : void

GuiView

- sceneRouter : SceneRouter
- lobbyModel : LobbyModel
- gameModel : GameModel

+ **GuiView**(lobbyModel : LobbyModel)
+ **setSceneRouter**(sceneRouter : SceneRouter) : void
+ **setGameModel**(gameModel : GameModel) : void
- **refreshGameIfActive**() : void

reads

reads

reads

reads

<< abstract >>
AbstractClientView

+ **onError**(message : String) : void

<< interface >>
ClientView

+ **onUsernameResult**(username : String, accepted : boolean, reason : String) : void
+ **onAvailableLobbiesUpdated**(lobbies : List<LobbyInfo>) : void
+ **onCurrentLobbyUpdated**(lobby : LobbyInfo) : void
+ **onShowAvailableTotems**() : void
+ **onLobbyDissolved**() : void
+ **onGameStarted**(gameId : String) : void
+ **onGameSetupCompleted**(turnOrder : List<String>, initialFood : Map<String, Integer>, board : BoardSnapshot) : void
+ **onPhaseChanged**(phase : PhaseType, currentPlayer : String, resolutionOrder : List<String>) : void
+ **onCurrentPlayerChanged**(nextPlayer : String) : void
+ **onTurnOrderEstablished**(turnOrder : List<String>) : void
+ **onTotemPlaced**(nickname : String, tileID : char) : void
+ **onTotemReturned**(nickname : String, turnOrderPosition : int) : void
+ **onOfferTilesUpdated**(offerTiles : List<OfferTileInfo>) : void
+ **onBoardUpdated**(newUpperRow : List<String>, newLowerRow : List<String>, deckRemainingCount : int) : void
+ **onEraChanged**(newEra : Era, newUpperRowBuildings : List<String>, newLowerRowBuildings : List<String>) : void
+ **onPlayerLimitsInitialized**(nickname : String, remainingUpper : int, remainingLower : int) : void
+ **onPlayerLimitsUpdated**(nickname : String, remainingUpper : int, remainingLower : int) : void
+ **onPlayerResourceChanged**(nickname : String, resource : ResourceType, newValue : int) : void
+ **onCardTaken**(nickname : String, cardID : String, cardType : CardType, sourceRow : RowPosition) : void
+ **onEventResolved**(EventID : String, eventName : String) : void
+ **onExtraTurnStarted**(nickname : String, remainingUpper : int, remainingLower : int) : void
+ **onExtraTurnEnded**(nickname : String) : void
+ **onGameEnded**(winners : List<String>, finalRankings : List<PlayerFinalScore>) : void
+ **onPlayerDisconnected**(nickname : String) : void
+ **onGameAborted**(lastManStanding : String) : void
+ **onGameRecoveryFailed**() : void
+ **onPlayerReconnected**(nickname : String) : void
+ **onError**(message : String) : void
+ **setGameModel**(gameModel : GameModel) : void
+ **onReturnToLobby**() : void
+ **onConnectionLost**() : void
+ **onConnectionRestored**() : void
+ **onAutoPlayerTimerStarted**(nickname : String, seconds : long) : void
+ **onAutoPlayerInvoked**(nickname : String) : void
+ **onGlobalTimerStarted**(seconds : long) : void
+ **onGlobalTimerCancelled**() : void

LobbyModel

- myNickname : String
- availableLobbies : List<LobbyInfo>
- currentLobby : LobbyInfo
- clientViews : List<ClientView>

+ **addObserver**(observer : ClientView) : void
+ **removeObserver**(observer : ClientView) : void
+ **updateUsernameResult**(username : String, isValid : boolean, reason : String) : void
+ **updateAvailableLobbies**(lobbies : List<LobbyInfo>) : void
+ **updateCurrentLobby**(lobby : LobbyInfo) : void
+ **updateLobbyDissolved**() : void
+ **updateError**(errorMessage : String) : void
+ **getMyNickname**() : String
+ **getAvailableLobbies**() : List<LobbyInfo>
+ **getCurrentLobby**() : LobbyInfo

GameModel

- myNickname : String
- gameId : String
- currentPhase : PhaseType
- currentPlayer : String
- totemByPlayer : Map<String, Totem>
- currentEra : Era
- turnOrder : List<String>
- upperRow : List<String>
- lowerRow : List<String>
- upperRowBuildings : List<String>
- lowerRowBuildings : List<String>
- offerTiles : List<OfferTileInfo>
- deckRemainingCount : int
- foodByPlayer : Map<String, Integer>
- ppByPlayer : Map<String, Integer>
- totemPositions : Map<String, Character>
- turnOrderPositions : Map<String, Integer>
- remainingUpper : Map<String, Integer>
- remainingLower : Map<String, Integer>
- charactersByPlayer : Map<String, List<String>>
- buildingsByPlayer : Map<String, List<String>>
- winners : List<String>
- finalRankings : List<PlayerFinalScore>
- gameEnded : boolean
- lastErrorMessage : String
- lastEventResolved : String
- turnOrderSlots : List<TurnOrderSlotInfo>
- clientViews : List<ClientView>

+ **GameModel**(myNickname : String)
+ **addObserver**(observer : ClientView) : void
+ **removeObserver**(clientView : ClientView) : void
+ **updateGameStarted**(gameID : String) : void
+ **updateGameSetupCompleted**(totemByPlayer : Map<String, Totem>, turnOrder : List<String>, initialFood : Map<String, Integer>, board : BoardSnapshot) : void
+ **updateCurrentPhase**(phase : PhaseType, currentPlayer : String, resolutionOrder : List<String>) : void
+ **updateCurrentPlayer**(nextPlayer : String) : void
+ **updateTurnOrder**(turnOrder : List<String>) : void
+ **updateTotemPlaced**(nickname : String, tileID : char) : void
+ **updateTotemReturned**(nickname : String, turnOrderPosition : int) : void
+ **updateBoard**(newUpperRow : List<String>, newLowerRow : List<String>, deckRemainingCount : int) : void
+ **updateEra**(newEra : Era, newUpperRowBuildings : List<String>, newLowerRowBuildings : List<String>) : void
+ **updatePlayerLimitsInitialized**(nickname : String, remainingUpper : int, remainingLower : int) : void
+ **updatePlayerLimits**(nickname : String, remainingUpper : int, remainingLower : int) : void
+ **updatePlayerResource**(nickname : String, resource : ResourceType, newValue : int) : void
+ **updateCardTaken**(nickname : String, cardID : String, cardType : CardType, sourceRow : RowPosition) : void
+ **updateAutoPlayerTimerStarted**(nickname : String, seconds : long) : void
+ **updateAutoPlayerInvoked**(nickname : String) : void
+ **updateEventResolved**(eventID : String, eventName : String) : void
+ **updateExtraTurnStarted**(nickname : String, remainingUpper : int, remainingLower : int) : void
+ **updateExtraTurnEnded**(nickname : String) : void
+ **updateGameEnded**(winners : List<String>, finalRankings : List<PlayerFinalScore>) : void
+ **updatePlayerDisconnected**(nickname : String) : void
+ **updateGameAborted**(lastManStanding : String) : void
+ **updateGameRecoveryFailed**() : void
+ **updatePlayerReconnected**(nickname : String) : void
+ **updateError**(errorMessage : String) : void
+ **updateGlobalTimerStarted**(seconds : long) : void
+ **updateGlobalTimerCancelled**() : void
+ **getMyNickname**() : String
+ **getGameId**() : String
+ **getCurrentPhase**() : PhaseType
+ **getCurrentPlayer**() : String
+ **getCurrentEra**() : Era
+ **getTurnOrder**() : List<String>
+ **getUpperRow**() : List<String>
+ **getLowerRow**() : List<String>
+ **getUpperRowBuildings**() : List<String>
+ **getLowerRowBuildings**() : List<String>
+ **getOfferTiles**() : List<OfferTileInfo>
+ **getFoodByPlayer**() : Map<String, Integer>
+ **getPpByPlayer**() : Map<String, Integer>
+ **getTotemPositions**() : Map<String, Character>
+ **getTurnOrderPositions**() : Map<String, Integer>
+ **getRemainingUpper**() : Map<String, Integer>
+ **getRemainingLower**() : Map<String, Integer>
+ **getCharactersByPlayer**() : Map<String, List<String>>
+ **getBuildingsByPlayer**() : Map<String, List<String>>
+ **getWinners**() : List<String>
+ **getFinalRankings**() : List<PlayerFinalScore>
+ **getDeckRemainingCount**() : int
+ **isGameEnded**() : boolean
+ **isInGame**() : boolean
+ **getLastEventResolved**() : String
+ **getLastErrorMessage**() : String
+ **getTurnOrderSlots**() : List<TurnOrderSlotInfo>
+ **getTotem**(nickname : String) : Totem
+ **setGameId**(gameId : String) : void

SceneRouter

- primaryStage : Stage
- guiController : GuiController
- rootContainer : StackPane
- baseLayer : StackPane
- modalLayer : StackPane
- toastLayer : VBox
- lobbyListScene : LobbyListScene
- lobbyScene : LobbyScene
- gameScene : GameScene

+ **SceneRouter**(primaryStage : Stage)
+ **setGuiController**(guiController : GuiController) : void
+ **show**() : void
+ **showProjectInfoScene**(onExit : Runnable) : void
+ **showIntroScene**() : void
+ **showGameMenuScene**() : void
+ **showLobbyListScene**(lobbies : List<LobbyInfo>) : void
+ **showLobbyScene**(currentLobby : LobbyInfo) : void
+ **switchToGameScene**(gameModel : GameModel) : void
+ **getLobbyListScene**() : LobbyListScene
+ **getLobbyScene**() : LobbyScene
+ **getGameScene**() : GameScene
+ **showToast**(title : String, content : String, type : AlertType) : void
+ **showBlockingAlert**(title : String, content : String, type : AlertType) : void
+ **showGameOverPopup**(title : String, content : String) : void
+ **showInGameMenu**() : void
+ **showSummaryCard**() : void
+ **hideModal**() : void
+ **showPlayerDisconnectedPopup**(n : String) : void
+ **showConnectionLost**() : void
+ **promptConnectionAndRetry**(lobbyModel : LobbyModel, clientView : ClientView) : void
- **initWindowArchitecture**() : void
- **runOnUI**(action : Runnable) : void
- **switchView**(newView : Region) : void

TuiController

- tuiView : TuiView
- networkType : NetworkType
- host : String
- port : int

+ **TuiController**(proxy : ServerProxy, lobbyModel : LobbyModel, view : ClientView, networkType : NetworkType, host : String, port : int)
+ **run**() : void
createNewProxy() : ServerProxy {throws Exception}
- **dispatch**(input : String) : void
- **parseNumericArg**(input : String, usage : String) : int

GuiController

- networkType : NetworkType
- host : String
- port : int

+ **GuiController**(proxy : ServerProxy, lobbyModel : LobbyModel, view : ClientView)
+ **setConnectionConfig**(networkType : NetworkType, host : String, port : int) : void
createNewProxy() : ServerProxy {throws Exception}
+ **getAvailableLobbies**() : List<LobbyInfo>
+ **requestSetUsername**(nickname : String) : void
+ **requestCreateLobby**(size : int) : void
+ **requestJoinLobby**(lobbyId : String) : void
+ **requestReconnect**(nick : String, gId : String) : void
+ **requestSelectTotem(t : Totem) : void**
+ **requestLeaveLobby**() : void
+ **moveTotem**(t : char) : void
+ **resolveActions**(ids : List<String>) : void
+ **requestReturnToLobby**() : void

<<abstract>>
ClientController

proxy : ServerProxy
lobbyModel : LobbyModel
view : ClientView
gameModel : GameModel

+ **ClientController**(proxy : ServerProxy, lobbyModel : LobbyModel, view : ClientView)
createNewProxy() : ServerProxy {throws Exception} {abstract}
+ **setServerProxy**(proxy : ServerProxy) : void
+ **setGameModel**(gameModel : GameModel) : void
+ **onGameModelRequired**(nickname : String) : void
performReturnToLobby() : void
+ **handleSetNickname**(nickname : String) : void
+ **handleCreateLobby**(size : int) : void
+ **handleJoinLobby**(index : int) : void
+ **handleSelectTotem**(totem : Totem) : void
+ **handleMoveTotem**(tileID : char) : void
+ **handleResolveActions**(cardIDs : List<String>) : void
+ **handleReconnect**(nickname : String, gameId : String) : void
+ **getGameModel**() : GameModel
myNickname() : String

<< interface >>
VirtualServer

+ **requestSetUsername**(username : String) : void
+ **requestCreateLobby**(numPlayers : int) : void
+ **requestJoinLobby**(lobbyID : String) : void
+ **requestSelectTotem**(color : Totem) : void
+ **requestLeaveLobby**() : void
+ **requestReconnect**(nickname : String, gameId : String) : void
+ **moveTotem**(tileID : char) : void
+ **resolveActions**(selectedIDs : List<String>) : void

<< interface >>
ServerProxy

+ **connect**() : void {throws Exception}
+ **disconnect**() : void
+ **isConnected**() : boolean
+ **setClientController**(controller : ClientController) : void

<< interface >> (sealed)
Command

+ **apply**(manager : VirtualControllerManager, clientId : String) : void

encodes

SocketServerProxy

- PING_INTERVAL_MILLIS : long
- PING_TIMEOUT_MILLIS : long
- host : String
- port : int
- lobbyModel : LobbyModel
- view : ClientView
- codec : JsonMessageCodec
- clientController : ClientController
- dispatcher : ClientNetworkDispatcher
- socket : Socket
- out : PrintWriter
- connected : boolean
- attemptingReconnection : boolean
- intentionalDisconnect : boolean
- pingThread : Thread
- lastPongReceivedAt : long
- activeNickname : String
- activeGameId : String

+ **SocketServerProxy**(host : String, port : int, lobbyModel : LobbyModel, view : ClientView)
+ **setClientController**(controller : ClientController) : void
+ **connect**() : void {throws Exception}
+ **disconnect**() : void
+ **isConnected**() : boolean
- **listenForEvents**() : void
- **startPingThread**() : void
- **stopPingThread**() : void
- **handleConnectionLost**() : void
- **send**(command : Command) : void

decodes / dispatch

<< interface >> (sealed)
Event

+ **apply**(view : VirtualView) : void

apply - IoC

<< interface >>
VirtualView

+ **notifyUsernameResult**(username : String, isValid : boolean, reason : String) : void
+ **notifyGameStarted**(gameId : String) : void
+ **notifyAvailableLobbiesUpdated**(lobbies : List<LobbyInfo>) : void
+ **notifyCurrentLobbyUpdated**(lobby : LobbyInfo) : void
+ **notifyLobbyDissolved**(lobbyID : String) : void
+ **notifyGameSetupCompleted**(totemByPlayer : Map<String, Totem>, turnOrder : List<String>, initialFood : Map<String, Integer>, boardSnapshot : BoardSnapshot) : void
+ **notifyPhaseChanged**(phase : PhaseType, currentPlayer : String, resolutionOrder : List<String>) : void
+ **notifyCurrentPlayerChanged**(nextPlayer : String) : void
+ **notifyTurnOrderEstablished**(turnOrder : List<String>) : void
+ **notifyBoardUpdated**(newUpperRow : List<String>, newLowerRow : List<String>, discardedCards : List<String>, movedToLowerRow : List<String>, deckRemainingCount : int) : void
+ **notifyEraChanged**(newEra : Era, newUpperRowBuildings : List<String>, newLowerRowBuildings : List<String>, discardedBuildings : List<String>) : void
+ **notifyTotemPlaced**(nickname : String, tileID : char) : void
+ **notifyTotemReturned**(nickname : String, turnOrderPosition : int) : void
+ **notifyCardTaken**(nickname : String, cardID : String, cardType : CardType, sourceRow : RowPosition) : void
+ **notifyPlayerLimitsInitialized**(nickname : String, remainingUpper : int, remainingLower : int) : void
+ **notifyPlayerLimitsUpdated**(nickname : String, remainingUpper : int, remainingLower : int) : void
+ **notifyPlayerResourceChanged**(nickname : String, resource : ResourceType, newValue : int, delta : int) : void
+ **notifyEventResolved**(eventID : String, eventName : String) : void
+ **notifyExtraTurnStarted**(nickname : String, remainingUpper : int, remainingLower : int) : void
+ **notifyExtraTurnEnded**(nickname : String) : void
+ **notifyGameEnded**(winners : List<String>, finalRankings : List<PlayerFinalScore>) : void
+ **notifyError**(message : String) : void
+ **notifyPlayerDisconnected**(nickname : String) : void
+ **notifyPlayerReconnected**(nickname : String) : void
+ **notifyAutoPlayerTimerStarted**(nickname : String, seconds : long) : void
+ **notifyAutoPlayerInvoked**(nickname : String) : void
+ **notifyGameAborted**(winner : String) : void
+ **notifyGameRecoveryFailed**() : void
+ **notifyGlobalTimerStarted**(seconds : long) : void
+ **notifyGlobalTimerCancelled**() : void

ClientNetworkDispatcher

- clientController : ClientController
- lobbyModel : LobbyModel
- onGameStarted : Consumer<String>
- activeNickname : String
- activeGameId : String

+ **ClientNetworkDispatcher**(clientController : ClientController, lobbyModel : LobbyModel)
+ **setOnGameStarted**(callback : Consumer<String>) : void
+ **updateActiveNickname**(nickname : String) : void
+ **updateActiveGameId**(gameId : String) : void
+ **updateInternalState**(nick : String, gId : String) : void
- **ensureGameModel**() : void

<<factory>>
ServerProxyFactory

- **ServerProxyFactory**()
+ **create**(type : NetworkType, host : String, port : int, lobbyModel : LobbyModel, view : ClientView) : ServerProxy {throws Exception}

creates

creates

RmiServerProxy (extends UnicastRemoteObject)

- POLL_TIMEOUT_SECONDS : long
- PING_INTERVAL_MILLIS : long
- outboundQueue : BlockingQueue<RemoteAction>
- outboundWorker : Thread
- pingThread : Thread
- running : boolean
- disconnected : AtomicBoolean
- host : String
- port : int
- lobbyModel : LobbyModel
- clientView : ClientView
- serverStub : RmiServerRemote
- clientController : ClientController
- dispatcher : ClientNetworkDispatcher
- connected : boolean
- attemptingReconnection : boolean
- intentionalDisconnect : boolean
- activeNickname : String
- activeGameId : String

+ **RmiServerProxy**(host : String, port : int, lobbyModel : LobbyModel, clientView : ClientView) {throws RemoteException}
+ **setClientController**(controller : ClientController) : void
+ **connect**() : void {throws Exception}
+ **disconnect**() : void
+ **isConnected**() : boolean
- **drainOutboundQueue**() : void
- **enqueue**(action : RemoteAction) : void
- **startPingThread**() : void
- **stopPingThread**() : void
- **handleNetworkFailure**(e : Exception) : void

stub

<< interface >> (extends Remote)
«<ghost>»
RmiServerRemote

<< interface >> (extends Remote)
RmiClientRemote

+ **notifyUsernameResult**(username : String, isValid : boolean, reason : String) : void {throws RemoteException}
+ **notifyGameStarted**(gameId : String) : void {throws RemoteException}
+ **notifyAvailableLobbiesUpdated**(lobbies : List<LobbyInfo>) : void {throws RemoteException}
+ **notifyCurrentLobbyUpdated**(lobby : LobbyInfo) : void {throws RemoteException}
+ **notifyLobbyDissolved**(lobbyID : String) : void {throws RemoteException}
+ **notifyGameSetupCompleted**(totemByPlayer : Map<String, Totem>, turnOrder : List<String>, initialFood : Map<String, Integer>, boardSnapshot : BoardSnapshot) : void {throws RemoteException}
+ **notifyPhaseChanged**(phase : PhaseType, currentPlayer : String, resolutionOrder : List<String>) : void {throws RemoteException}
+ **notifyCurrentPlayerChanged**(nextPlayer : String) : void {throws RemoteException}
+ **notifyTurnOrderEstablished**(turnOrder : List<String>) : void {throws RemoteException}
+ **notifyBoardUpdated**(newUpperRow : List<String>, newLowerRow : List<String>, discardedCards : List<String>, movedToLowerRow : List<String>, deckRemainingCount : int) : void {throws RemoteException}
+ **notifyEraChanged**(newEra : Era, newUpperRowBuildings : List<String>, newLowerRowBuildings : List<String>, discardedBuildings : List<String>) : void {throws RemoteException}
+ **notifyTotemPlaced**(nickname : String, tileID : char) : void {throws RemoteException}
+ **notifyTotemReturned**(nickname : String, turnOrderPosition : int) : void {throws RemoteException}
+ **notifyCardTaken**(nickname : String, cardID : String, cardType : CardType, sourceRow : RowPosition) : void {throws RemoteException}
+ **notifyPlayerLimitsInitialized**(nickname : String, remainingUpper : int, remainingLower : int) : void {throws RemoteException}
+ **notifyPlayerLimitsUpdated**(nickname : String, remainingUpper : int, remainingLower : int) : void {throws RemoteException}
+ **notifyPlayerResourceChanged**(nickname : String, resource : ResourceType, newValue : int, delta : int) : void {throws RemoteException}
+ **notifyEventResolved**(eventID : String, eventName : String) : void {throws RemoteException}
+ **notifyExtraTurnStarted**(nickname : String, remainingUpper : int, remainingLower : int) : void {throws RemoteException}
+ **notifyExtraTurnEnded**(nickname : String) : void {throws RemoteException}
+ **notifyGameEnded**(winners : List<String>, finalRankings : List<PlayerFinalScore>) : void {throws RemoteException}
+ **notifyError**(message : String) : void {throws RemoteException}
+ **notifyPlayerDisconnected**(nickname : String) : void {throws RemoteException}
+ **notifyPlayerReconnected**(nickname : String) : void {throws RemoteException}
+ **notifyAutoPlayerTimerStarted**(nickname : String, seconds : long) : void {throws RemoteException}
+ **notifyAutoPlayerInvoked**(nickname : String) : void {throws RemoteException}
+ **notifyGameAborted**(winner : String) : void {throws RemoteException}
+ **notifyGameRecoveryFailed**() : void {throws RemoteException}
+ **notifyGlobalTimerStarted**(seconds : long) : void {throws RemoteException}
+ **notifyGlobalTimerCancelled**() : void {throws RemoteException}
+ **ping**() : void {throws RemoteException}